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Heterogeneous Multiagent Task Allocation via Cooperative Exchange Strategies for Equilibrium-Improving: A Potential Game Framework

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***Abstract*—Task allocation in multiagent systems is a critical challenge due to the heterogeneity of tasks and agents, where tasks have varying resource requirements and agents possess differing resource supplies. During execution, agents' resources deplete while tasks' requirements dynamically decrease. This problem has broad applications, such as optimally deploying UAVs equipped with diverse medical supplies in disaster rescue scenarios to minimize casualties. To address this, this article proposes a coalition formation game model, formulated as a potential game. We theoretically prove the submodularity of both the coalition utility function and the global utility function. Based on this submodularity, we establish that the efficiency lower bound of any Nash equilibrium in the proposed game model is given by $e/(2e-1)$, significantly outperforming the 50% bound reported in prior studies. Furthermore, we introduce an inertia-based log-linear learning algorithm enhanced with a multiagent cooperative exchange mechanism, which enables the system to escape from suboptimal equilibria and improve global utility. In addition, we extend the algorithm to accommodate local communication constraints and dynamic allocation scenarios. Extensive experimental evaluations demonstrate that our proposed method achieves superior performance across diverse scenarios compared to existing algorithms.**

***Index Terms*—Log-linear learning, multiagent system, potential game, task allocation.**

I. Introduction

MULTIAGENT systems enable agents to work together and solve complex real-world problems that individual agents cannot handle alone [1]. Such cooperative mechanisms have found widespread applications in various domains [2], [3]. Examples include disaster response operations [4], [5], military missions [6], industrial manufacturing [7], mobile crowdsensing [8], and many others [9], [10], [11]. Among core challenges in multiagent systems, task allocation plays a crucial role in determining the overall system performance. An efficient task allocation mechanism ensures maximizing the system's efficiency, which constitutes a fundamental problem called the multiagent task allocation (MATA) problem.

In this article, we concentrate on the single-task, multiagent, instantaneous allocation (ST-MA-IA) problem [13], where a task can be assigned to multiple agents, while each agent can only select at most one task at a time. In addition, the nature of the problem discussed in this article introduces several critical characteristics as follows.

1) *Tasks and Agents Are Heterogeneous:* Tasks require diverse types and quantities of resources, while agents possess different types and quantities of resources. This heterogeneity calls for an allocation mechanism that adapts to varied capability-task matches for efficient resource utilization.
2) *Agents' Resources Are Consumable:* Agents have limited, consumable resources used in task execution. For instance, in a rescue scenario, consumable resources may include medical supplies and food. The consumable nature of resources highlights the need for efficient task allocation strategies to accomplish more tasks with limited capacity.
3) *Agents' Decision-Making Is Distributed:* Agents make decisions in a distributed manner under a communication structure represented by an undirected, connected graph $\mathcal{G}$. Each agent communicates locally with its neighbors, and full communication among agents is achieved when $\mathcal{G}$ is a complete graph. Distributed decision-making improves scalability and robustness in large-scale systems but also poses greater challenges for designing local coordination mechanisms under varying communication topologies.
4) *The Task Allocation Process Is Dynamic:* Given that resources are consumable, if agents and tasks remain after the initial allocation, a subsequent allocation process is triggered. The dynamic nature of task allocation requires continuous decision adjustments to adapt to changing tasks and resource availability.

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Generally, distributed methods to the task allocation problem are typically classified into three categories: 1) market-based; 2) hedonic game-based; and 3) potential game-based.

Market-based methods have been widely applied to task allocation across various domains. In UAV swarms, the probability-tuned market algorithm (PTMA) handles target uncertainty and task-ordering constraints [14], while decentralized auction algorithms ensure conflict-free assignments in dynamic autonomous vehicle networks [15]. Simulations further demonstrate the robustness of market-based strategies in wireless sensor networks [16]. Hedonic game-based approaches model agent preferences and use distributed algorithms to form stable coalitions. For homogeneous swarms, decentralized methods perform effectively under local information constraints [17], [18], and for heterogeneous swarms, Wang et al. [19] extended the framework to enable capability-aware coalition formation. Saad et al. [20] further applied hedonic games to wireless data-gathering agents, analyzing convergence and stability. However, market-based methods often rely on centralized pricing mechanisms, while hedonic game-based methods, although ensuring coalition stability, typically involve a tradeoff with overall system efficiency.

Potential games provide a distributed framework where agents maximizing their own utilities lead to the optimization of system efficiency. Wu and Shang [1] developed a potential game framework for MATA with heterogeneous agents. Yang et al. [21] addressed heterogeneous-cost task allocation with budget constraints through a potential game and proposed an algorithm, referred to in this article as the time variant log-linear learning with cooperative exchange strategy (TVLL-CE). Specifically, this framework has also been applied to the task assignment problem in UAV swarms [22], [23], [24] and task assignment and power allocation problems in radar networks [25], [26]. However, these studies differ from ours in several aspects: agents in [1] and [24] possess nonconsumable abilities; UAV swarms in [23] are homogeneous; coalition formation in [22] is overlapping, and in [21] it is restricted to a single stage; and radar node locations in [25] and [26] remain fixed.

The MATA problem that incorporates all four aforementioned characteristics poses substantial complexity, and existing studies have not yet provided a comprehensive modeling framework. Therefore, this article proposes a potential game-theoretic task allocation model designed to accommodate these complexities.

Distributed algorithms for computing the Nash equilibrium of potential games have been extensively studied. Among them, iterative best response-based algorithms, such as joint strategy fictitious play (JSFP) [27], the best response algorithm (BRA) [28], and the better reply process (BRP) [29], are widely adopted due to their guaranteed convergence and ease of distributed implementation. However, their greedy update mechanism, which always selects the best-response strategy, may converge to suboptimal equilibria, especially in nonconvex potential functions. In contrast, log-linear learning-based algorithms (LLAs) introduce stochastic updates that occasionally accept suboptimal strategies, thereby enhancing exploration of the strategy space. Representative algorithms following this approach include time-variant log-linear learning (TVLL) [30], the binary log-linear learning algorithm (BLLA) [31], and the inertia log-linear learning algorithm (ILLA). However, these algorithms may still suffer from convergence to suboptimal equilibria due to uncertainty in the update process [21].

The above analysis highlights the importance of designing algorithms that can effectively escape local optima. Furthermore, ensuring that the proposed algorithm can operate under local communication constraints is also of critical importance.

Motivated by the aforementioned content, two problems are considered in this article.

1) How to construct a game-theoretic distributed task allocation framework that accommodates the proposed characteristics and validates its effectiveness?
2) How to design an algorithm that can escape from locally optimal Nash equilibria to achieve higher global utility, and is applicable to local communication scenarios?

To address the first problem, we formulate a potential game that supports distributed decision-making. The utility function captures the heterogeneity and consumability of resources through resource-task matching and prioritization of high-value tasks. Moreover, the dynamic allocation process is captured by iteratively executing the coalition formation game (CFG) until resources are depleted or tasks are completed. To validate the effectiveness of the proposed model, submodularity has been widely used in recent studies [1], [17] to improve scalability and computational efficiency of multiagent systems. We demonstrate that the global utility function is submodular, thereby establishing a performance bound for any Nash equilibrium and confirming the model's effectiveness.

To address the second challenge, we incorporate the multiagent cooperative exchange (MACE) strategy into the ILLA to escape suboptimal Nash equilibria and enhance global utility. Specifically, we further extend the algorithm to support local communication scenarios.

The main contributions of this article are outlined as follows.

1) This article proposes a distributed task allocation framework based on potential game, where the coalition utility and global utility are proven to be submodular. It is theoretically proven that our game model guarantees at least $(e/(2e-1))$ of the optimum, surpassing the 50% approximation bound for any Nash equilibrium established in [1] and [17]. Moreover, it is proved that the optimal solution itself is also a Nash equilibrium of our model.
2) This article proposes the ILLA with MACE (ILLA-MACE), which enables strategy exchange among multiple agents and reduces the computational cost associated with traversal via an inertia mechanism, in contrast to the cooperative exchange (CE) strategy [21]. We further extend ILLA-MACE to local communication settings via a timestamp-based protocol and adapt it to dynamic environments.
3) Through extensive numerical experiments, we demonstrate the effectiveness of the proposed method across various scenarios. Compared with existing algorithms [1], [17], [21], [27], [28], [29], [30], it consistently achieves superior performance under both resource-abundant and resource-constrained conditions. Its performance in local communication and dynamic task allocation settings further underscores the scalability of our approach.

The remainder of this article is organized as follows. Section II presents the problem formulation. Section III introduces the model and related definitions. Section IV analyzes the Nash equilibrium of the game model. Section V presents algorithms for various scenarios. Section VI presents simulation results, and Section VII concludes this article.

II. Problem Formulation

Consider a task allocation scenario where tasks require distinct types and quantities of resources. Heterogeneous agents, each equipped with distinct types and amounts of resources, form coalitions to collaboratively accomplish these tasks. Each agent's resources are consumable, and the required resources of tasks gradually diminish during execution. An agent cannot perform multiple tasks at the same time, yet each task can be executed by several agents.

Tasks: Suppose there exist a set of $m+1$ tasks denoted by $T = \{t_1, t_2, \ldots, t_m\} \cup \{t_\emptyset\}$, where $t_\emptyset$ is the void task. Namely, when $t_\emptyset$ is allocated to some agents, they do not perform any actions. Each task $t_j \in T$ is described by the attribute tuple $\Theta_{t_j} = \langle \boldsymbol{L}_j, \boldsymbol{p}_{t_j}, V_j \rangle$. Here, $\boldsymbol{L}_j = [l_j^1, l_j^2, \ldots, l_j^K]$ denotes the resources requirements of task t_j, with K being the total number of resource types. $\boldsymbol{p}_{t_j}$ represents the position of task t_j, and V_j denotes its value.

Agents: Consider a group of n agents denoted as the set $A = \{a_1, a_2, \ldots, a_n\}$. Each agent a_i is characterized by an attribute tuple $\Theta_{a_i} = \langle \boldsymbol{c}_i, \boldsymbol{p}_{a_i}, r_i \rangle$. The vector $\boldsymbol{c}_i = [c_i^1, c_i^2, \ldots, c_i^K]$ denotes the resources carried by agent a_i. The vector $\boldsymbol{p}_{a_i}$ is the position of agent a_i. During the task execution, agents need to communicate with each other, and r_i is the communication radius of agent a_i. The definition of communication networks is given in Definition 1.

Definition 1 (Communication Networks): Information interactions among the agents can be represented by an undirected connected graph $\mathcal{G} = (A, \xi, H)$, where A is the vertex set, $\xi \subset A \times A$ is the edge set, and $H = [h_{ij}] \in \mathbb{R}^{N\times N}$ is the associated adjacency matrix. Here, the symbol $\times$ denotes the Cartesian product. An edge between agent a_i and a_j is denoted by (a_i, a_j), which means the information exchange between them. The associated adjacency matrix H is given by

$$h_{ij} = \begin{cases} 1, & \text{if } (a_i, a_j) \in \xi \\ 0, & \text{else.} \end{cases} \tag{1}$$

The neighbor of agent a_i is defined as a set of agents within its communication radius r_i, which is given by

$$N_i = \{a_j \in A | \, \|\boldsymbol{p}_{a_i} - \boldsymbol{p}_{a_j}\| \le r_i\} \tag{2}$$

where $|N_i|$ refers to the number of neighbors of agent a_i, and $\|\boldsymbol{p}_{a_i} - \boldsymbol{p}_{a_j}\|$ denotes the Euclidean distance (i.e., the L_2 norm) between a_i and a_j.

Let the vector $\boldsymbol{x}_i = [x_{i1}, \ldots, x_{im}]$ be the strategy of agent a_i, where $x_{ij} = 1$ if task t_j is assigned to agent a_i, and $x_{ij} = 0$ otherwise. Then, the MATA problem can be described as the following optimization problem.

Problem 1: The MATA problem consists a set of n agents $A = \{a_1, a_2, \ldots, a_n\}$ and a set of $m+1$ tasks $T = \{t_1, t_2, \ldots, t_m\} \cup \{t_\emptyset\}$. The optimization objective of this problem can be defined as follows:

$$\max_{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n} \mathrm{UT} \tag{3}$$

$$\text{s.t.} \sum_{t_j \in T} x_{ij} \le 1 \quad \forall a_i \in A \tag{4}$$

$$x_{ij} \in \{0, 1\} \quad \forall a_i \in A \; \forall t_j \in T \tag{5}$$

where UT is the global utility.

The optimization objective of Problem 1 is to maximize the global utility by optimizing the task allocation of each agent. During the task allocation process, agents performing the same task form coalitions and agents without executable tasks form a virtual coalition. The definition of coalition structure is given as follows:

Definition 2 (Coalition Structure): A coalition structure or a coalition partition is defined as the set $\pi = \{s_1, \ldots, s_m, s_{m+1}\}$ which partitions the agent set A into disjoint coalitions such that $\cup_{j=1}^{m+1} s_j = A$. For $\forall j = 1, 2, \ldots, m$, s_j denotes the coalition of agents that select task t_j, and s_{m+1} denotes the virtual coalition formed by agents that select $t_\emptyset$.

III. Coalition Formation Game

In this section, we propose a CFG and prove it is a potential game, thereby transforming it into a distributed optimization problem for the agents.

A. Reward Utility Function

The reward utility function for coalition s_j reflects the rewards obtained from task execution, which is determined by both the coalition's task completion rate (CR) and the task's priority. It is defined as follows:

$$R(s_j) = \gamma_j \lambda (e^{-G(s_j)} - 1) \tag{6}$$

where γ_j is the priority of task r_j denoted as $\gamma_j = \left(V_j / \sum_{j'=1}^{m} V_{j'}\right)$, and $\lambda = (e/1-e)$ is a negative constant that ensures $R(s_j) \in [0, \gamma_j]$. $G(s_j)$ is the average task CR of coalition s_j, which is given as follows:

$$G(s_j) = \frac{\sum_{k=1}^{K} P_j^k}{\|\boldsymbol{L}_j\|_0} \tag{7}$$

where $\|\boldsymbol{L}_j\|_0$ denotes the L_0 norm of $\boldsymbol{L}_j$, which counts the number of nonzero elements in the resource demand vector of task t_j. P_j^k is the resource satisfaction rate of kth resource of given coalition s_j, which is defined as follows:

$$P_j^k = \begin{cases} 0, & \text{if } l_j^k = 0 \\ \min\left\{1, \dfrac{\sum_{i \in s_j} c_i^k}{l_j^k}\right\}, & \text{otherwise.} \end{cases} \tag{8}$$

In (6), $R(s_j)$ is a monotonically increasing function of $G(s_j)$. When $G(s_j) = 1$, $R(s_j)$ reaches its maximum value of γ_j, indicating that the coalition's resources meet the task's requirements. Conversely, when $G(s_j) = 0$, $R(s_j)$ takes the minimum value of 0, meaning the coalition lacks the resources needed for the task.

B. Cost Function

Task execution imposes time and energy costs on agents, both of which are related to the distance. Thus, we define a distance-based cost function for coalition s_j

$$C(s_j) = \sum_{i \in s_j} C_{ij} \tag{9}$$

where C_{ij} is the distance cost of agent a_i related to task t_j, which is motivated from [13] and given by

$$C_{ij} = 1 - \gamma_j d_{ij} \tag{10}$$

where d_{ij} is the normalized distance between agent a_i and task t_j, which belongs to $[0, 1]$.

C. Utility Function

The utility function evaluates the suitability of coalition members for a task. In this article, the utility function of coalition s_j is defined as follows [13]:

$$U(s_j) = R(s_j) + \rho C(s_j). \tag{11}$$

Following the work in [25], we adopt $\rho \in [0, \rho_{\max}]$ as a constant cost factor. Its value can be adjusted for different scenarios: setting $\rho = 0$ when agents need to accomplish tasks at all costs (e.g., in emergency rescue), and selecting a larger ρ to reduce costs when resources are abundant.

Accordingly, we can derive the global utility function UT as follows:

$$\mathrm{UT} = \sum_{s_j \in \pi} U(s_j). \tag{12}$$

It is worth noting that both the monotonic increase of cost C_{ij} in (10) as distance decreases and the reward-plus-cost operation in (11) may appear counterintuitive. However, these designs ensure (9) and (11) remain nondecreasing, which is a necessary condition for proving the submodularity of the coalition utility function in Theorem 2.

Definition 3 (CFG Model): The CFG model Γ is a tuple $\langle A, \Pi, \mathrm{UT}\rangle$, where: 1) $A = \{a_1, a_2, \ldots, a_n\}$ denotes a set of n agents; 2) Π denotes the set of all possible coalition structures; and 3) UT denotes the global utility function of the agent set, defined in (12).

To formulate the MATA problem as a distributed optimization framework, we will prove that the proposed CFG model constitutes a *potential game*. To satisfy the potential game condition, each agent's utility $U_i(s_j)$ is designed as its marginal contribution to coalition s_j according to the Wonderful-Life Utility formalism [32]

$$U_i(s_j) = \begin{cases} U(s_j) - U(s_j \setminus \{a_i\}), & \text{if } a_i \in s_j \\ U(s_j \cup \{a_i\}) - U(s_j), & \text{if } a_i \notin s_j. \end{cases} \tag{13}$$

Specifically, $s_j \setminus \{a_i\}$ means the coalition s_j with agent a_i removed. The definition of a potential game is as follows.

Definition 4 (Potential Game): A game $G\langle A, \{\boldsymbol{x}_i, i \in A\}, \{U_i, i \in A\}\rangle$ is called a potential game if there exists a global function $\Phi : \boldsymbol{x}_i \to \boldsymbol{R}, \forall i \in A, \boldsymbol{x}_i, \boldsymbol{x}_i' \in \boldsymbol{X}_i, \boldsymbol{x}_{-i} \in \boldsymbol{X}_{-i}$, such that

$$U_i(\boldsymbol{x}_i, \boldsymbol{x}_{-i}) - U_i(\boldsymbol{x}_i', \boldsymbol{x}_{-i}) = \Phi(\boldsymbol{x}_i, \boldsymbol{x}_{-i}) - \Phi(\boldsymbol{x}_i', \boldsymbol{x}_{-i}) \tag{14}$$

where the global function Φ is known as the potential function of game G.

Then, we will prove that the proposed game (CFG) model is a potential game, which serves as the cornerstone for subsequent equilibrium analysis.

Theorem 1: The CFG is a potential game with global utility UT defined in (12) as the potential function.

Proof: It is assumed that $a_i \in s_j$ for all $a_i \in A$. Based on (13), it holds that

$$U_i(\boldsymbol{x}_i, \boldsymbol{x}_{-i}) = U\left(s_j\right) - U\left(s_j \setminus \{a_i\}\right)$$

where $\boldsymbol{x}_{-i}$ refers to the strategies of other agents apart from agent a_i. If a_i switches from coalition s_j to $s_{j'}$, its utility is given by

$$U_i\left(\boldsymbol{x}_i', \boldsymbol{x}_{-i}\right) = U\left(s_{j'} \cup \{a_i\}\right) - U\left(s_{j'}\right).$$

Hence, the change of individual utility is given as follows:

$$\begin{aligned} &U_i(\boldsymbol{x}_i, \boldsymbol{x}_{-i}) - U_i(\boldsymbol{x}_i', \boldsymbol{x}_{-i}) \\ &\quad = U(s_j) - U(s_j \setminus \{a_i\}) - U(s_{j'} \cup \{a_i\}) + U(s_{j'}). \end{aligned} \tag{15}$$

For the global utility UT, when $a_i \in s_j$, it follows that

$$\begin{aligned} \mathrm{UT}(\boldsymbol{x}_i, \boldsymbol{x}_{-i}) = U(s_1) + \cdots + U\left(s_j\right) + \cdots \\ + U\left(s_{j'}\right) + \cdots + U\left(s_{m+1}\right). \end{aligned}$$

When $a_i \in s_{j'}$, the global utility UT can be given by

$$\begin{aligned} \mathrm{UT}\left(\boldsymbol{x}_i', \boldsymbol{x}_{-i}\right) = U(s_1) + \cdots + U\left(s_j \setminus \{a_i\}\right) + \cdots \\ + U\left(s_{j'} \cup \{a_i\}\right) + \cdots + U\left(s_{m+1}\right). \end{aligned}$$

So we have

$$\begin{aligned} &\mathrm{UT}(\boldsymbol{x}_i, \boldsymbol{x}_{-i}) - \mathrm{UT}(\boldsymbol{x}_i', \boldsymbol{x}_{-i}) \\ &\quad = U(s_j) - U(s_j \setminus \{a_i\}) + U(s_{j'}) - U(s_{j'} \cup \{a_i\}). \end{aligned} \tag{16}$$

Combining (15) and (16), it holds that $U_i(\boldsymbol{x}_i, \boldsymbol{x}_{-i}) - U_i(\boldsymbol{x}_i', \boldsymbol{x}_{-i}) = \mathrm{UT}(\boldsymbol{x}_i, \boldsymbol{x}_{-i}) - \mathrm{UT}(\boldsymbol{x}_i', \boldsymbol{x}_{-i})$. According to Definition 4, the proof is completed.■

Theoretically, this result ensures the existence of a Nash equilibrium for the proposed game. Practically, it enables self-organizing systems where local decisions naturally converge to an equilibrium.

IV. Nash Equilibrium Analysis

In this section, we demonstrate the submodularity of both coalition utility functions and global utility functions, and derive a suboptimal lower bound for any Nash equilibrium.

Definition 5 (Nash Equilibrium): A strategy profile $\boldsymbol{x}^* = \{\boldsymbol{x}_1^*, \ldots, \boldsymbol{x}_n^*\}$ is a Nash equilibrium if and only if no agent can improve its individual utility by deviating from its strategy unilaterally while other agents keep theirs unchanged, i.e.,

$$U_i(\boldsymbol{x}_i^*, \boldsymbol{x}_{-i}^*) \geq U_i(\boldsymbol{x}_i, \boldsymbol{x}_{-i}^*) \quad \forall \boldsymbol{x}_i \in \boldsymbol{X}, \boldsymbol{x}_i \neq \boldsymbol{x}_i^*. \tag{17}$$

Definition 6 (Submodular Function): A set function $f : A \to R$ is submodular if one of the following two conditions is satisfied.

1) $f(X) + f(Y) \geq f(X \cap Y) + f(X \cup Y), \forall X, Y \subseteq V$.
2) $f(X \cup \{x\}) - f(X) \geq f(Y \cup \{x\}) - f(Y), \forall X, Y \subseteq V, X \subseteq Y, \forall x \in V \setminus Y$.

A set function f is nondecreasing if $f(X) \leq f(Y)$ for any $X \subseteq Y \subseteq V$.

Next, we analyze the optimality of Nash equilibria in our model and establish its effectiveness by leveraging the submodularity of the utility function.

Theorem 2: The coalition utility function $U(s_j)$ defined in (11) is a submodular set function.

Proof: By letting $\alpha_1 = \gamma_j \lambda < 0$ in (6), we obtain $R(s_j) = \alpha_1(e^{-G(s_j)} - 1)$. For coalition s_j, let $\beta_j = (1/\|\boldsymbol{L}_j\|_0)$. According to (7), we have $G(s_j) = \beta_j \sum_{k=1}^{K} P_j^k$.

Without loss of generality, among the K required resource types, it is assumed that the first I types can be satisfied by coalition s_j, i.e., $P_j^1, \ldots, P_j^I = 1$, while the resources from type $I+1$ to q are invalid, i.e., $P_j^{I+1}, \ldots, P_j^q = 0$

$$R(s_j) = \alpha_1 \left(e^{-\beta_j (I + P_j^{q+1} + \cdots + P_j^K)} - 1 \right). \quad (18)$$

Let $\alpha_2 = e^{-\beta_j I}$ and $\mu_j^k = \beta_j / l_j^k$, then

$$R(s_j) = \alpha_1 \left(\alpha_2 e^{-\sum_{k=q+1}^K \sum_{i \in s_j} \mu_j^k c_i^k} - 1 \right).$$

For $\forall X, Y \subseteq V, X \subseteq Y, x \in V \setminus Y$, let $g(X) = \sum_{k=q+1}^K \sum_{i \in X} \mu^k c_i^k$ and $A(\{x\}) = \sum_{k=1}^K \mu^k c_x^k$, we have the following equations about the reward utility function:

$$R(X \cup \{x\}) - R(X) = \alpha_1 \alpha_2 e^{-g(X)} (e^{-A(\{x\})} - 1) \quad (19)$$

$$R(Y \cup \{x\}) - R(Y) = \alpha_1 \alpha_2 e^{-g(Y)} (e^{-A(\{x\})} - 1). \quad (20)$$

For the cost function, we have

$$C(X) = |X| - \gamma_j \sum_{i \in X} d_{ij} \quad (21)$$

$$C(X \cup \{x\}) = |X| + 1 - \gamma_j \left(\sum_{i \in X} d_{ij} + d_{xj} \right). \quad (22)$$

Similarly, it can be deduced that $C(X \cup \{x\}) - C(X) = C(Y \cup \{x\}) - C(Y) = 1 - \gamma_j d_{xj}$. Combining the above equations, we have

$$\begin{aligned} &U(X \cup \{x\}) - U(X) \\ &\quad = \alpha_1 \alpha_2 e^{-g(X)} (e^{-A(\{x\})} - 1) + \rho(1 - \gamma_j d_{xj}) \quad (23) \\ &U(Y \cup \{x\}) - U(Y) \\ &\quad = \alpha_1 \alpha_2 e^{-g(Y)} (e^{-A(\{x\})} - 1) + \rho(1 - \gamma_j d_{xj}). \quad (24) \end{aligned}$$

As $g(\cdot)$ refers to represents the average CR of the coalition for task types from $q+1$ to K, and $X \subseteq Y$, we have $g(X) \leq g(Y)$. Obviously, $\alpha_1 \alpha_2 (e^{-A(\{x\})} - 1) > 0$ and $e^{-g(X)} \geq e^{-g(Y)}$. Therefore

$$U(X \cup \{x\}) - U(X) \geq U(Y \cup \{x\}) - U(Y). \quad (25)$$

Hence, the coalition utility function $U(s_j)$ is submodular.■

Based on the results of Theorem 2, Theorem 3 further proves the submodularity of global utility.

Theorem 3: The global utility $\mathrm{UT} = \sum_{s_j \in \pi} R(s_j)$ defined in (12) is a submodular set function.

Proof: Let $Z_i = \{z_i^1, z_i^2, \ldots, z_i^m\}$ be the action set of agent a_i, where z_i^j denotes the action that agent a_i selects task t_j. So the action set of the entire agent group can be denoted by $\mathcal{Z} = Z_1 \cup Z_2 \cup \cdots \cup Z_m$. Then the global utility is a function UT: $2^{\mathcal{Z}} \to \mathbb{R}$, where $2^{\mathcal{Z}}$ is the power set of $\mathcal{Z}$. Each element in $2^{\mathcal{Z}}$ induces a coalition structure. We denote this mapping by a function CS: $2^{\mathcal{Z}} \to \Pi$. Since each agent can only choose at most one task, not every subset of $2^{\mathcal{Z}}$ qualifies as a feasible action profile. Toward this end, let Ω be the set of all feasible action profiles.

For $\forall X, Y \subseteq \Omega, X \subseteq Y, \forall x \in \Omega \setminus Y$, let π_X and π_Y be the corresponding coalition structure of X and Y, and $\pi_X(t_j)$ refers to the coalition executing task t_j under the action profile X. Given that $X \subseteq Y$, it holds that $\pi_X(t_j) \subseteq \pi_Y(t_j), \forall t_j \in \mathcal{T}$. Let $t' \in \mathcal{T}$ be the task related to the action x and a' be the corresponding agent, then we have

$$\mathrm{UT}(X \cup \{x\}) - \mathrm{UT}(X) = U(\pi_X(t') \cup \{a'\}) - U(\pi_X(t')) \quad (26)$$

$$\mathrm{UT}(Y \cup \{x\}) - \mathrm{UT}(Y) = U(\pi_Y(t') \cup \{a'\}) - U(\pi_Y(t')). \quad (27)$$

Given the above analysis, two cases arise based on task t'.

1) If $\pi_X(t') = \pi_Y(t')$, it holds that $U(\pi_X(t') \cup \{a'\}) - U(\pi_X(t')) = U(\pi_Y(t') \cup \{a'\}) - U(\pi_Y(t'))$, then we have $\mathrm{UT}(X \cup \{x\}) - \mathrm{UT}(X) = \mathrm{UT}(Y \cup \{x\}) - \mathrm{UT}(Y)$.
2) If $\pi_X(t') \subset \pi_Y(t')$, according to Theorem 2, $U(\pi_X(t') \cup \{a'\}) - U(\pi_X(t')) \geq U(\pi_Y(t') \cup \{a'\}) - U(\pi_Y(t'))$, then we have $\mathrm{UT}(X \cup \{x\}) - \mathrm{UT}(X) \geq \mathrm{UT}(Y \cup \{x\}) - \mathrm{UT}(Y)$.

In summary, it holds that

$$\mathrm{UT}(X \cup \{x\}) - \mathrm{UT}(X) \geq \mathrm{UT}(Y \cup \{x\}) - \mathrm{UT}(Y). \quad (28)$$

Hence, the global utility UT is a submodular set function.■

Submodularity, which implies diminishing marginal gains for expanding a coalition, is a key property for ensuring that the utility function promotes effective coalition formation.

By combining the results of Theorems 1–3, the optimality of the Nash equilibria is analyzed in Theorem 4.

Theorem 4: Consider a game $G = \{N, \{A_i\}, \mathrm{UT}\}$, where UT is defined in (12). Then any equilibrium s^{ne} satisfies $(e/(2e-1))\mathrm{UT}(s^{\mathrm{opt}}) \leq \mathrm{UT}(s^{\mathrm{ne}}) \leq \mathrm{UT}(s^{\mathrm{opt}})$, where s^{opt} is the optimal strategy that has the maximum global utility. Moreover, there exists an equilibrium $\hat{s}^{\mathrm{ne}}$ such that $\mathrm{UT}(s^{\mathrm{opt}}) = \mathrm{UT}(\hat{s}^{\mathrm{ne}})$.

Proof: For the nondecreasing and submodular set function UT, [33, Th. 5] implies that for any Nash equilibrium s^{ne} it holds

$$\frac{1}{1 + \delta(\mathrm{UT})} \mathrm{UT}(s^{\mathrm{opt}}) \leq \mathrm{UT}(s^{\mathrm{ne}}) \leq \mathrm{UT}(s^{\mathrm{opt}}). \quad (29)$$

Based on the definition of discrete curvature in [33], $\delta(\mathrm{UT})$ is given by

$$\delta(\mathrm{UT}) = 1 - \min_{\substack{x \in \Omega, \\ A = \Omega \setminus \{x\}, \\ \mathrm{UT}(x) > 0}} \left(\frac{\mathrm{UT}(A \cup \{x\}) - \mathrm{UT}(A)}{\mathrm{UT}(\{x\})} \right). \quad (30)$$

According to (26) and (27), let $\hat{A} = \pi_A(t')$, and a' is the agent corresponding to action x. We have

$$\mathrm{UT}(A \cup \{x\}) - \mathrm{UT}(A) = U(\hat{A} \cup \{a'\}) - U(\hat{A}) \quad (31)$$

$$\mathrm{UT}(\{x\}) = U(\pi_x(t')) = U(\{x\}). \quad (32)$$

The reward function of coalition X is a set function $R(X)$, which can be denoted by

$$R(X) = \alpha_1 \left[e^{-\sum_{k=1}^K \sum_{i \in X} \eta_i^k c_i^k} - 1 \right] \quad (33)$$

where

$$\eta_i^k = \begin{cases} 0, & \text{if } l_j^k = 0, \\ \beta, & \text{if } l_j^k \neq 0 \text{ and } \sum_{i \in X} c_i^k > l_j^k, \\ \dfrac{\beta}{l^k}, & \text{if } l_j^k \neq 0 \text{ and } \sum_{i \in X} c_i^k \leq l_j^k. \end{cases}$$

Hence, we have the following.

1) $R(\hat{A}) = \alpha_1 \left[e^{-\sum_{k=1}^K \sum_{i \in \hat{A}} \eta_i^k c_i^k} - 1 \right]$.
2) $R(\hat{A} \cup \{x\}) = \alpha_1 \left[e^{-\sum_{k=1}^K \sum_{i \in \hat{A}} \eta_i^k c_i^k - \sum_{k=1}^K \eta_x^k c_x^k} - 1 \right]$.
3) $R(\{x\}) = \alpha_1 \left[e^{-\sum_{k=1}^K \eta_x^k c_x^k} - 1 \right]$.

Similarly, the cost function is given by the following.

1) $C(\hat{A}) = |\hat{A}| - \gamma_j \sum_{i \in \hat{A}} d_{ij}$.
2) $C(\hat{A} \cup \{x\}) = |\hat{A}| + 1 - \gamma_j \left(\sum_{i \in \hat{A}} d_{ij} + d_{xj} \right)$.
3) $C(\{x\}) = 1 - \gamma_j d_{xj}$.

Hence, we have

$$U(\hat{A}\cup\{a'\})-U(\hat{A}) = \alpha_1 e^{-\sum_{k=1}^{K}\sum_{i\in\hat{A}}\eta_i^k c_i^k}\left(e^{-\sum_{k=1}^{K}\eta_{a'}^k c_{a'}^k}-1\right) + \rho(1-\gamma_j d_{a'j}) \tag{34}$$

$$U(\{a'\}) = \alpha_1\left(e^{-\sum_{k=1}^{K}\eta_{a'}^k c_{a'}^k}-1\right)+\rho(1-\gamma_j d_{a'j}). \tag{35}$$

Let $P_{\hat{A}}=\sum_{k=1}^{K}\sum_{i\in\hat{A}}\eta_i^k c_i^k$ and $P_{a'}=\sum_{k=1}^{K}\eta_{a'}^k c_{a'}^k$, denoted by the average CR of coalition $\hat{A}$ and $\{a'\}$ for task t', respectively. Obviously, $P_{\hat{A}}, P_{a'}\in[0,1]$. Given (31), we have

$$\frac{\mathrm{UT}(A\cup\{x\})-\mathrm{UT}(A)}{\mathrm{UT}(\{x\})} = \frac{\alpha_1 e^{-P_{\hat{A}}}(e^{-P_{a'}}-1)+\rho(1-\gamma_j d_{a'j})}{\alpha_1(e^{-P_{a'}}-1)+\rho(1-\gamma_j d_{a'j})}. \tag{36}$$

Specifically

$$\frac{\alpha_1 e^{-P_{\hat{A}}}(e^{-P_{a'}}-1)+\rho(1-\gamma_j d_{a'j})}{\alpha_1(e^{-P_{a'}}-1)+\rho(1-\gamma_j d_{a'j})}-\frac{\alpha_1 e^{-P_{\hat{A}}}(e^{-P_{a'}}-1)}{\alpha_1(e^{-P_{a'}}-1)} = \frac{\rho(1-\gamma_j d_{a'j})(1-e^{-P_{\hat{A}}})}{\alpha_1(e^{-P_{a'}}-1)+\rho(1-\gamma_j d_{a'j})}. \tag{37}$$

Obviously, $1-e^{-P_{\hat{A}}}\geq 0$, $\alpha_1(e^{-P_{a'}}-1)\geq 0$, and $\rho(1-\gamma_j d_{a'j})\geq 0$. Therefore, it holds that

$$\frac{\mathrm{UT}(A\cup\{x\})-\mathrm{UT}(A)}{\mathrm{UT}(\{x\})} \geq \frac{\alpha_1 e^{-P_{\hat{A}}}(e^{-P_{a'}}-1)}{\alpha_1(e^{-P_{a'}}-1)} = e^{-P_{\hat{A}}}\geq\frac{1}{e}. \tag{38}$$

Combining with (36)–(38), it holds that

$$\frac{\mathrm{UT}(A\cup\{x\})-\mathrm{UT}(A)}{\mathrm{UT}(\{x\})}\geq\frac{1}{e}. \tag{39}$$

Based on (29) and (30), we have $\delta(\mathrm{UT})\leq 1-(1/e)$. Therefore, it holds that $\mathrm{UT}(s^{\mathrm{ne}})\geq(1/(1+\delta(\mathrm{UT})))\mathrm{UT}(s^{\mathrm{opt}})\geq(e/(2e-1))\mathrm{UT}(s^{\mathrm{opt}})$.

Suppose, for contradiction, that s^{opt} is not a Nash equilibrium. Then, there exists at least one agent a_i that can increase its utility by deviating from $\boldsymbol{x}_i$ to $\boldsymbol{x}_i^*$, i.e., $U_i(\boldsymbol{x}_i^*,\boldsymbol{x}_{-i}) > U_i(\boldsymbol{x}_i,\boldsymbol{x}_{-i})$. In the potential game, this means that the global utility can be improved by changing the strategy of agent a_i from $\boldsymbol{x}_i$ to $\boldsymbol{x}_i^*$. Obviously, this result contradicts the assumption. Therefore, strategy s^{opt} is a Nash equilibrium.■

Theorem 4 theoretically proves the effectiveness of the proposed model through the suboptimal lower-bound analysis of an arbitrary Nash equilibrium.

V. Algorithm Design

In this section, we introduce the ILLA-MACE algorithm to solve for the Nash equilibrium, and extend it to the local communication and dynamic task allocation scenarios.

A. ILLA-MACE Algorithm

In order to solve potential games, several algorithms [27], [28], [29], [34], [35] demonstrate effective convergence performance. Specifically, LLA guarantees that the action profile converges to a potential maximizer in potential games [36]. The probability of agent x choosing action a_x of the log-linear learning dynamics is given as follows:

$$p_x^{a_x}(t)=\frac{e^{\beta U_x(a_x,a_{-x}(t-1))}}{\sum_{\bar{a}_x\in A_x}e^{\beta U_x(\bar{a}_x,a_{-x}(t-1))}} \tag{40}$$

where the coefficient $\beta\geq 0$ controls the agent's tendency to choose actions.

However, LLA struggles to guarantee convergence to the optimal Nash equilibrium within a finite time [37], leading to many variants of LLA.

1) *TVLL:* The probability of agent x choosing action a_x is determined by

$$p_x^{a_x}(t)=\frac{e^{\beta(t)U_x(a_x,a_{-x}(t-1))}}{\sum_{\bar{a}_x\in A_x}e^{\beta(t)U_x(\bar{a}_x,a_{-x}(t-1))}} \tag{41}$$

where $\beta(t)$ is a time variant parameter given by

$$\beta(t)=\beta_0+\frac{\ln(\lambda t+1)}{c} \tag{42}$$

where $t\geq 0$, $\beta_0\geq 0$, $\lambda\geq 1$, $c\in\mathbb{N}^*$.

2) *BLLA:* This algorithm calculates the utility of only two actions at a time, which can reduce computational complexity

$$p_x^{a_x}(t)=\frac{e^{\beta U_x(a_x,a_{-x}(t-1))}}{e^{\beta U_x(a_x,a_{-x}(t-1))}+e^{\beta U_x(a_x(t-1),a_{-x}(t-1))}}. \tag{43}$$

3) *ILLA:* In this algorithm, each agent decides whether to change its action according to an inertia value, which is more general compared to the asynchronous rules (only one agent is selected at each time step) and the synchronous rules (all agents are selected at each time step).

Although existing algorithms converge to potential game equilibria, these may be suboptimal. A CE strategy [21] is proposed to improve utility through pairwise exchanges. In this article, we extend this strategy to the MACE strategy.

Definition 7 (MACE Strategy): Let $A=\{a_1,\ldots,a_i,\ldots,a_{i+p},\ldots,a_n\}$ be the set of agents and $T=\{t_1,\ldots,t_m,t_{m+1}\}$ be the set of tasks. For arbitrary p agents $a_i,\ldots,a_{i+p}\in A$ and their respective tasks $t(a_i),\ldots,t(a_{i+p})\in T$, where $t(a_i)$ denotes the task assigned to agent a_i. Let $\hat{T}=\{t(a_i),\ldots,t(a_{i+p})\}$ and $\hat{T}'=\{t'(a_i),\ldots,t'(a_{i+p})\}$ be the task set of these agents before and after the exchange, respectively, and satisfy $\hat{T}'\subseteq\hat{T}$. Then, the agents prefer to exchange their tasks if and only if it holds

$$\mathrm{UT}\left(t(a_1),\ldots,t(a_i),\ldots,t\left(a_{i+p}\right),\ldots,t(a_n)\right) < \mathrm{UT}\left(t(a_1),\ldots,t'(a_i),\ldots,t'\left(a_{i+p}\right),\ldots,t(a_n)\right).$$

In comparison with the CE strategy, the MACE strategy enhances the efficiency of the Nash equilibrium in potential games by enabling coordinated deviations by multiple agents. To clarify the distinction between the MACE and CE strategies, we present an illustrative example in Fig. 1.

As shown in Fig. 1, there are three tasks and nine agents. A Nash equilibrium is represented by the coalition structure $\pi_1=\{\{a_1,a_2\},\{a_3\},\{a_4,a_5,a_6\},\{a_7,a_8,a_9\}\}$, which yields a global utility of 10. The coalition $\{a_7,a_8,a_9\}$ corresponds to a virtual coalition. Assuming that π_1 is a locally optimal Nash equilibrium, an alternative coalition structure

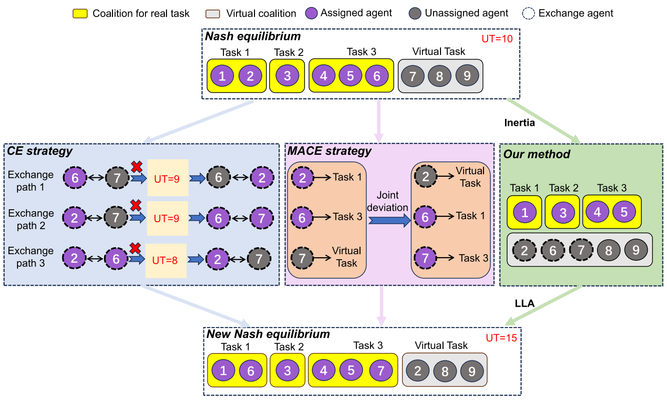

Fig. 1. Comparison of CE and MACE strategy.

$\pi_2 = \{\{a_1, a_6\}, \{a_3\}, \{a_4, a_5, a_7\}, \{a_2, a_8, a_9\}\}$ achieves a higher global utility of 15.

For the CE strategy, there are three exchange paths, each requiring two steps to reach the new coalition structure π_2. However, these paths are invalid because the first step would decrease the global utility.

For the MACE strategy, a_2, a_6, and a_7 can reach a new Nash equilibrium via a joint deviation. However, evaluating all possible joint strategy deviations across all coalition structures incurs a high computational cost in large-scale scenarios, as demonstrated by the algorithm based on the CE strategy [21]. In this article, we address this issue through an inertia-based selection mechanism that avoids the need for a full traversal. Based on this idea, we propose the inertia log-linear learning with MACE properties (ILLA-MACE) algorithm, as illustrated in the green section on the right side of Fig. 1, and whose pseudocode is shown in Algorithm 1.

Algorithm 1 ILLA-MACE Algorithm

1: **Input:** Parameters: maximum iterations $T_{\max}$, initial rate β_0, decay parameter λ, constant c, inertia r, seed s, error threshold ε, threshold of utility unchanged times f, boolean variable $flag$ is an indicator to identify whether the algorithm is operating in the MACE stage
2: **Initialize:** Empty task participants, initial agent assignments, $count \leftarrow 0$, $flag \leftarrow false$
3: **for** $t = 1$ **to** $T_{\max}$ **do**
4:     Calculate learning rate according to (42)
5:     $A_{\text{update}} \leftarrow \emptyset$
6:     **for** each agent i in A **do**
7:         **if** $UniformRandom(0, 1) > r$ **then**
8:             Append i to A_{update}
9:         **end if**
10:     **end for**
11:     **if** $count \geq f$ **then**
12:         **for** each agent i in A_{update} **do**
13:             Move agent i to virtual coalition
14:         **end for**
15:         $count \leftarrow 0$, $flag \leftarrow true$
16:     **end if**
17:     Record old global utility U_{old}
18:     **for** each agent i in A_{update} **do**
19:         Calculate utilities for all tasks
20:         Select task with log-linear probability $P(j) \propto e^{\beta_t U_{ij}}$
21:     **end for**
22:     **if** $flag = false$ or ($flag = true$ and $U_{\text{new}} > U_{\text{old}}$) **then**
23:         update the assignment
24:     **end if**
25:     calculate the new utility U_{new}
26:     **if** $|U_{\text{new}} - U_{\text{old}}| < \varepsilon$ **then**
27:         $count \leftarrow count + 1$
28:     **else**
29:         $count \leftarrow 0$
30:     **end if** $flag \leftarrow false$
31: **end for**
32: **Output:** Final coalition structure

Initially, agents whose uniformly generated random numbers exceed a predefined inertia parameter are included in the strategy update set A_{update} (lines 5–10). This inertia mechanism avoids the exhaustive traversal of all agents in the TVLL-CE algorithm, thereby reducing computational cost. When the global utility remains unchanged for f iterations, which suggests that the algorithm may have converged to a local optimum, and A_{update} are moved to the virtual coalition (lines 11–16). After recording the current global utility U_{old}, each agent in A_{update} updates its strategy according to the log-linear learning rule (lines 17–21). Then we determine whether to accept the new assignment based on the flag variable and the comparison between the new and old utility values (lines 22–24). Steps 11–24 correspond to the implementation of the MACE strategy. This process iterates until it reach $T_{\max}$, at which point the final coalition structure is output.

B. TB-ILLA-MACE Algorithm

In this section, inspired by [38], we propose the timestamp-based inertial log-linear algorithm with MACE properties (TB-ILLA-MACE), an extension of ILLA-MACE designed for local communication scenarios.

In full communication scenarios, agent a_i selects actions based on (40). In local communication scenarios, however, agents only observe limited information about others' actions.

Following [38], we define $\theta^i(t) = (z_i(t), \theta^i_{-i}(t))$ as the estimate of the joint action profile at time t, where z_i is the actual action of agent a_i and $\theta^i_{-i}(t) = (\theta^i_1(t), \theta^i_2(t), \ldots, \theta^i_{i-1}(t), \theta^i_{i+1}(t), \ldots, \theta^i_n(t))$ denote agent a_i's estimates of the actions of all other agents at time t. In local communication scenarios, the probability of agent a_i choosing action $z_i \in T$ is given by

$$p_i^{z_i}(t) = \frac{e^{\beta u_i(z_i, \theta^i_{-i}(t-1))}}{\sum_{z_i' \in T} e^{\beta u_i(z_i', \theta^i_{-i}(t-1))}}. \tag{44}$$

For $a_i \in N$, the timestamp vector $\sigma^i(t) = (t, \sigma^i_{-i}(t))$ records the generation time of each estimate corresponding to $\theta^i(t)$, which is used to achieve information consistency among different agents under a local communication scenario.

The detailed procedure of the TB-ILLA-MACE algorithm is presented in Algorithm 2. During each iteration, if an inertial condition is satisfied (line 4), agent a_i is selected to update its strategy. If agent a_i enters the MACE stage, it joins the virtual coalition and updates its action (lines 5 and 6). Otherwise, it calculates the utilities of all tasks based on the estimated

Algorithm 2 TB-ILLA-MACE Algorithm for Agent $a_i \in A$

1: **Input:** Maximum iterations $T_{\max}$, task set T, inertia r, neighboring agent set N_i
2: **Initialize:** Randomly select an initial action $\theta_i^i(0) \in T$, randomly initialize the estimate of action profile $\theta_{-i}^i \in T$ for all other agents, the timestamp variable $\sigma^i(0) = (0, 0, \ldots, 0)$, $t = 1$, boolean variable $flag = false$
3: **for** $t = 1$ **to** $T_{\max}$ **do**
4:     **if** $UniformRandom(0, 1) > r$ **then**
5:         **if** $flag = true$ **then**
6:             Move agent a_i to virtual coalition and update $z_i(t)$
7:         **else**
8:             Calculate utilities for all tasks based on $\theta^i(t-1)$
9:             Select action $z_i(t)$ using (44)
10:         **end if**
11:         Update $flag$ according to the change of global utility
12:         Receive messages $\Omega = \{\sigma^p(t), \theta^p(t)\}, p \in N_i$ from neighbors
13:         Update estimate and timestamp using (45) and (46)
14:         Send $\theta^i(t)$ and $\sigma^i(t)$ to all neighbors in N_i
15:     **end if**
16: **end for**
17: **Output:** Updated action profile and timestamps

actions of other agents and selects an action according to (44) (lines 7–9). The update method for flag follows lines 11–16 in Algorithm 1. Agent a_i receives messages from its neighbors, updates its estimates and timestamps accordingly, and then sends the updated estimates and timestamps to all neighbors (lines 12–14). The above process is repeated until $T_{\max}$ is reached.

When agents communicate with their neighbors, the update rules of the estimate and timestamp are given as follows:

$$\theta_j^i(t) = \theta_j^a(t-1) \tag{45}$$

$$\sigma_j^i(t) = \sigma_j^a(t-1) \tag{46}$$

where $a = \arg\max_{k \in N_i} \sigma_j^k(t-1)$.

C. Dynamic Task Assignment

Given that resources are consumable, if there are remaining agents and tasks after one allocation, a subsequent task allocation is initiated. In this section, we focus on the dynamic task allocation problem.

To execute a task, a coalition must consume the necessary resources, which are supplied by its members. If multiple members can provide the same resource type, resources are consumed from the agent with the smallest index first. The value of a task, denoted $V_j^{(t)}$ for task t_j at time t, is a function that decays based on the cumulative resources consumed. Its value at time $t+1$ is given by

$$V_j^{(t+1)} = V_j^{(t)} \left(1 - \sum_{k=1}^{K} P_j^k\right). \tag{47}$$

The flowchart in Fig. 2 depicts the dynamic task assignment process. Initially, the system inputs the attributes of tasks and agents, calculates task priorities, and computes coalition and individual utilities. Based on the communication structure, it determines the optimal coalition using Algorithm 1 for full communication or Algorithm 2 for local communication, allowing agents to proceed to their assigned tasks.

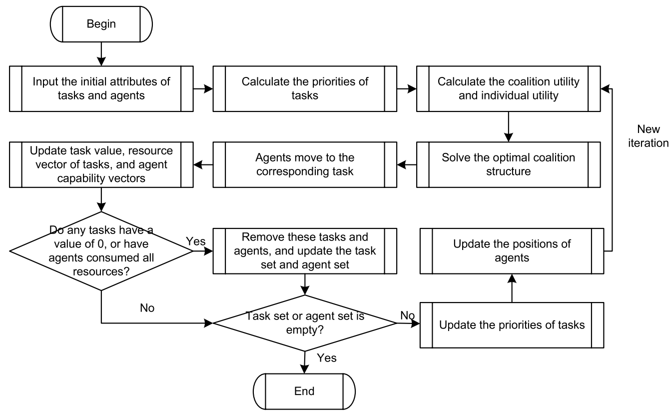

Fig. 2. Flowchart of dynamic task assignment.

D. Algorithm Comparison Analysis

In this section, since the TB-ILLA-MACE algorithm is an extension of ILLA-MACE, we focus on comparing and analyzing ILLA-MACE with several classical algorithms and provide the time complexity of both proposed algorithms.

ILLA-MACE is a variant of the log-linear learning framework and differs from iterative best-response methods, such as JSFP, BRA, and BRP. These methods typically perform greedy updates based on current best responses, which may converge quickly but get trapped in suboptimal equilibria.

Our algorithm combines the time-varying parameter from TVLL with the inertia mechanism from ILLA, and introduces the MACE strategy to escape local optima and enhance global exploration. Unlike TVLL-CE, which traverses all agents in each iteration, ILLA-MACE selectively identifies agents for coalition exchange based on inertia, reducing computational complexity. Moreover, by allowing multiple agents to update their strategies simultaneously, ILLA-MACE improves exploration to reach a better Nash equilibrium.

Given the number of agents n, the number of tasks m, the maximum number of iterations $T_{\max}$, and the inertia coefficient r, the time complexity of ILLA-MACE is $\mathcal{O}\big(T_{\max} \cdot rnm\big)$. The time complexity of TB-ILLA-MACE is $\mathcal{O}\big(T_{\max} \cdot (rnm + N_{\max})\big)$, where $N_{\max}$ denotes the maximum number of neighbors among all agents.

VI. Simulations and Analysis

In this section, we evaluate the performance of our approach through the following experiments: an optimality analysis of the equilibria (see VI-A), a comparison with other algorithms under varying resource quantities (see VI-B), an ablation experiment on the MACE strategy (see VI-C), an evaluation under local communication constraints (see VI-D), tests in dynamic task allocation scenarios (see VI-E), and a sensitivity analysis of the parameters (see VI-F).

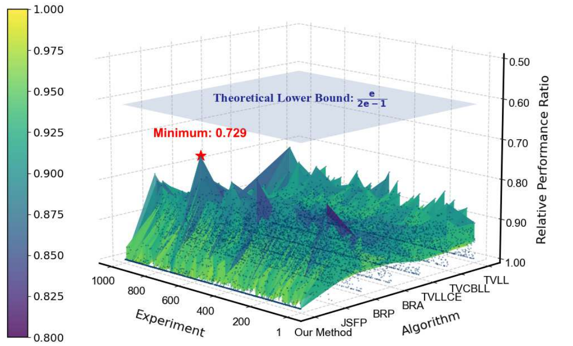

Fig. 3. Lower bound of Nash equilibrium verification. The x-axis and y-axis represent the experiment index and different algorithms, respectively, while the z-axis shows the ratio between the utility at the Nash equilibrium and the optimal utility, as indicated by the scatter points. The blue plane denotes the theoretical lower bound of this ratio. The colormap on the left reflects the magnitude of the performance ratio. For clarity, the 100 lowest points for each algorithm (700 points in total) are selected to form a triangulated surface, illustrating the variation in equilibrium quality across algorithms.

A. Optimality Analysis of Nash Equilibrium

First, we validate the suboptimality bound of any Nash equilibrium in the proposed model through Monte Carlo experiments.

We use the ILLA-MACE algorithm, conducting 1000 experiments, where in each run the number of agents is randomly sampled from $[8, 12]$, the number of tasks from $[2, 6]$, and the resource dimension from $[4, 8]$. Agents' resources and tasks' resource requirements are generated under three scenarios: resource-abundant, resource-scarce, and balanced. Each algorithm's performance is measured by the ratio of converged utility to the optimal utility (via exhaustive search).

Fig. 3 confirms that with our utility design, all algorithms consistently converge to solutions exceeding the theoretical lower bound. Notably, the JSFP algorithm achieves the lowest ratio of 0.729. These results validate the effectiveness of the proposed game model.

Then, by comparing the Nash equilibrium produced by ILLA-MACE and other algorithms with the optimal Nash equilibrium (via exhaustive search), we demonstrate the effectiveness and superiority of our algorithm.

We consider an urban task assignment problem involving three tasks and 12 agents. Each agent has four possible choices (including the virtual coalition), leading to a total of 4^{12} potential coalition structures.

By exhaustively traversing all possible coalition structures, the optimal coalition structure can be identified. However, such an approach requires about 700 s to execute on a personal computer equipped with a GeForce RTX 3060 Graphics Card.

Subsequently, ILLA-MACE, along with six existing algorithms, is applied to this scenario. As shown in Fig. 4, only our algorithm converges to the optimal global utility.

Table I compares different algorithms across various indicators, including global utility (UT), convergence iterations (Conv.), gap to optimal global utility (Gap), CR, and distance cost (Cost). ILLA-MACE yields the highest task CR and the lowest distance cost, while TVLL achieves the lowest global utility.

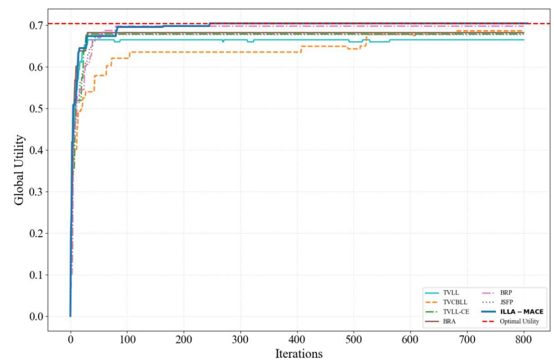

Fig. 4. Convergence curves of different algorithms.

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS

Algorithm	UT	Conv.	Gap (%)	CR (%)	Cost
TVLL	0.665	88	5.59	86.16	59.39
TVCBLL	0.686	105	2.55	87.84	58.45
TVLL-CE	0.679	**28**	3.63	88.02	61.31
BRA	0.682	30	3.13	88.37	60.37
BRP	0.697	80	0.97	88.36	57.31
JSFP	0.678	44	3.78	87.40	59.41
ILLA-MACE	**0.704**	246	**0.00**	**89.10**	**56.07**

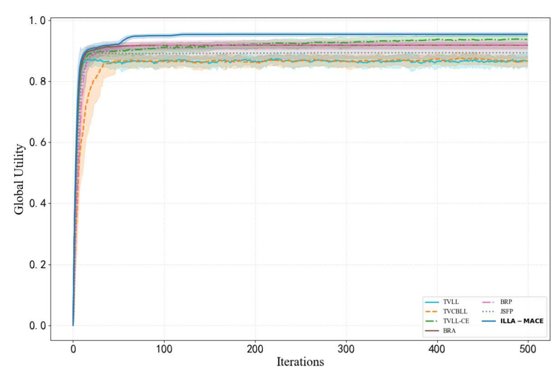

Fig. 5. Algorithm comparison under resource-abundant scenarios.

B. Algorithm Comparison Under Resource-Abundant and Resource-Constrained Scenario

This section considers an experimental setup with 20 agents and six tasks under full communication, evaluated in both resource-abundant and resource-constrained scenarios. We propose six performance indicators to assess the algorithms in these two scenarios: 1) average global utility (UT); 2) utility standard deviation (Std.); 3) average convergence iterations (Conv.); 4) average running time (Time); 5) average CR; and 6) average cost (Cost). Monte Carlo simulations with 1000 runs are conducted, with each run randomly sampling the attributes of agents and tasks.

1) Resource-Abundant Scenario: In resource-abundant scenarios, the resources provided by the agents far exceed the resources required by the tasks. In this section, we set $K = 5$, and for all $a_i \in A, t_j \in T, l_j^k, c_i^k \in [1, 10], k = 1, 2, \ldots, K$. Under

TABLE II
PERFORMANCE INDICATORS OF DIFFERENT ALGORITHMS UNDER RESOURCE-ABUNDANT SCENARIOS

Algorithm	UT	Std.	Conv.	Time	CR(%)	Cost
TVLL	0.870	0.020	1000.0	0.357	0.980	96.619
TVCBLL	0.867	0.016	1000.0	0.333	0.979	98.193
TVLL-CE	0.931	0.014	496.0	0.822	0.993	54.590
BRA	0.915	0.011	**52.0**	**0.267**	0.995	63.766
BRP	0.915	0.012	91.0	0.340	0.994	64.183
JSFP	0.897	0.013	80.0	0.347	0.996	80.295
ILLA-MACE	**0.952**	**0.002**	69.0	0.337	**0.999**	**37.923**

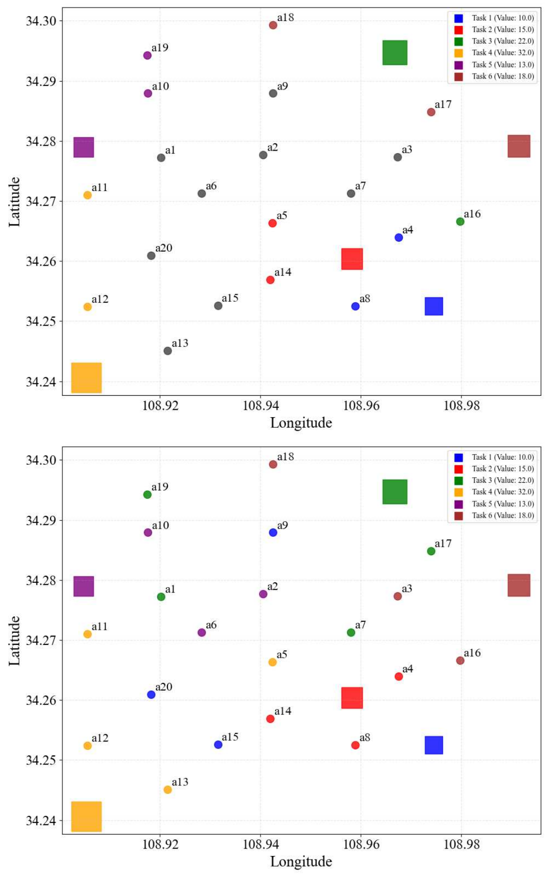

Fig. 6. Coalition structure outcomes under different resource scenarios. The top and bottom subfigures illustrate the coalition structure outcomes under resource-abundant and resource-constrained scenarios, respectively.

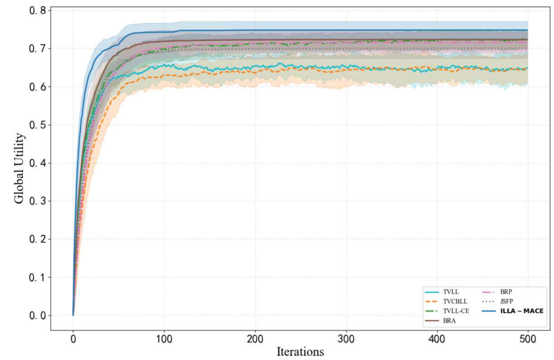

Fig. 7. Algorithm comparison under resource-constrained scenarios.

TABLE III
PERFORMANCE INDICATORS OF DIFFERENT ALGORITHMS UNDER RESOURCE-CONSTRAINED SCENARIOS

Algorithm	UT	Std.	Conv.	Time	CR(%)	Cost
TVLL	0.648	0.021	1000.0	0.343	0.682	114.343
TVCBLL	0.651	0.023	1000.0	0.309	0.685	114.335
TVLL-CE	0.712	0.015	495.0	0.817	0.718	95.036
BRA	0.709	0.010	**102.0**	**0.263**	0.721	98.109
BRP	0.702	0.012	148.0	0.654	0.361	103.990
JSFP	0.695	0.011	107.0	0.664	0.358	104.253
ILLA-MACE	**0.733**	**0.002**	103.0	0.305	**0.729**	**89.089**

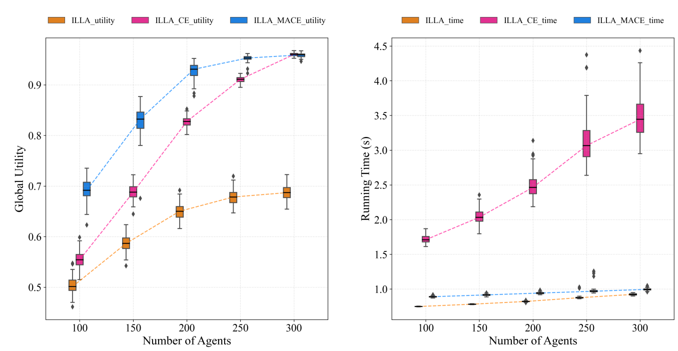

Fig. 8. Comparison of CE strategy and MACE strategy with fixed task quantity. The left subfigure shows that the global utility increases with the number of agents. The right subfigure shows the running time of algorithms with the increase of the number of agents.

this setting, the results of 1000 Monte Carlo simulations of seven algorithms are presented in Fig. 5 and Table II.

In Fig. 5, ILLA-MACE quickly converges to the optimal global utility. Conversely, TVLL and the time variant constrained binary log-linear learning (TVCBLL) [1] algorithm yield relatively low global utility under this scenario.

Table II shows that ILLA-MACE achieves the highest average global utility (0.952) and average CR (0.999), while exhibiting the lowest utility standard deviation (0.002) and average cost (37.923). TVLL and TVCBLL cannot converge within the maximum iteration limit.

Under this scenario, some agents tend to join the virtual coalition, as they cannot provide additional marginal contributions once the resource requirements have been satisfied. These agents are the gray circles in the top subfigure of Fig. 6.

2) Resource-Constrained Scenario: In resource-constrained scenarios, the resources provided by agents are far less than the resources required by tasks. In this section, we set $K = 5$, and for $\forall a_i \in A, t_j \in T, l_j^k \in [20, 30], c_i^k \in [1, 10], k = 1, 2, \ldots, K$. The simulation results for this scenario are presented in Fig. 7 and Table III.

In Fig. 7, ILLA-MACE exhibits outstanding performance, rapidly converging to the highest global utility.

In Table III, ILLA-MACE achieves the highest average global utility (0.733) and average CR (0.729), as well as the lowest utility standard deviation (0.002) and average cost (89.089).

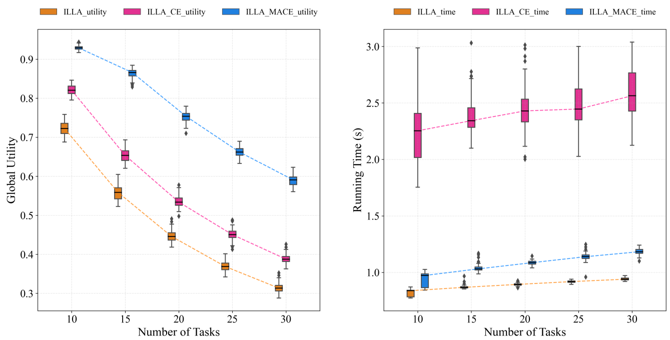

Fig. 9. Comparison of CE strategy and MACE strategy with fixed agent quantity. The left subfigure and right subfigure reflect the changes in global utility and running time, respectively, with an increasing number of tasks.

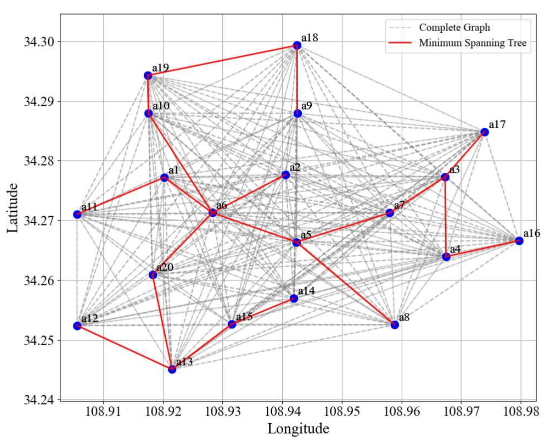

Fig. 10. Maximum and minimum communication radius. $r_{\min}$ corresponds to the minimum communication radius derived from the minimum spanning tree (MST) displayed in red lines, and $r_{\max}$ represents the communication radius in a fully connected graph displayed in gray dashed lines.

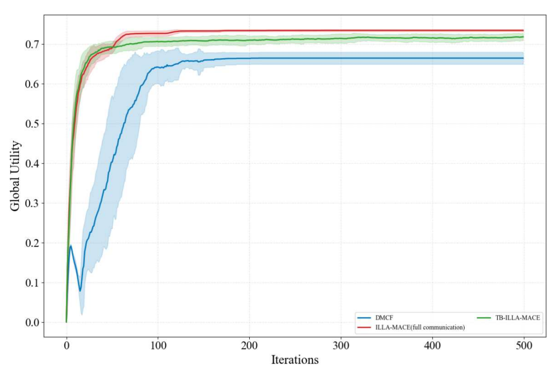

Fig. 11. Convergence comparison of DMCF and TB-ILLA-MACE algorithms.

Under this scenario, agents are partitioned into six coalitions displayed in the bottom subfigure of Fig. 6. No agents join the virtual coalition, indicating that all agents are fully utilized.

C. CE Strategy versus MACE Strategy

In this section, we conduct ablation experiments on the MACE strategy. Under the same initial conditions and scenario settings, we replace the agent policy update process based on the MACE strategy with one based on the CE strategy

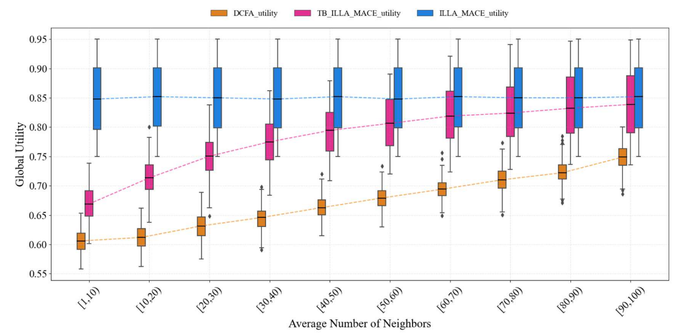

Fig. 12. Comparison of global utility across algorithms with varying average number of neighbors.

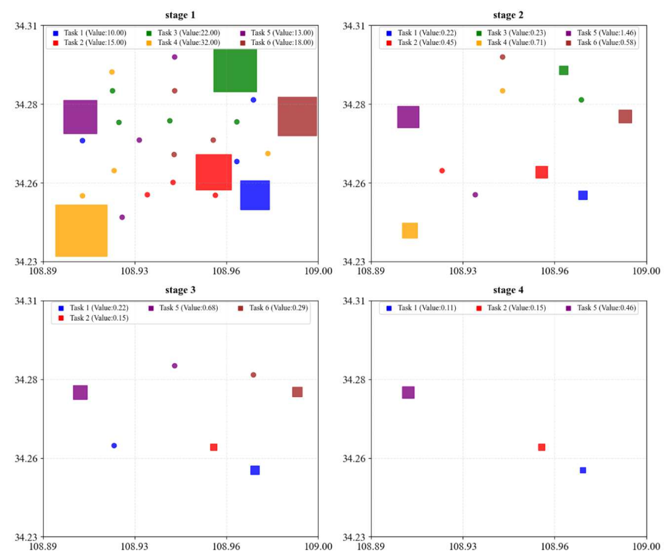

Fig. 13. Dynamic task assignment process. Each square and cycle represents a task and an agent, respectively. The size of the squares indicates the value of each task, while the colored circles represent agent coalitions that have chosen the same task. As the stages progress, agents consume their resources to execute the assigned tasks, leading to a reduction in both their available resources and the value of the tasks (indicated by the shrinking squares). When an agent exhausts all its resources, it will quit this process.

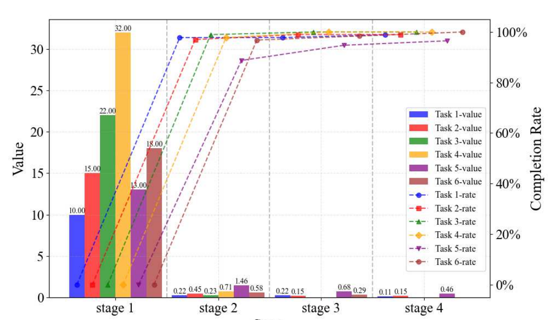

Fig. 14. Dynamic trend of task values and CRs. The bar chart displays the values of six tasks decreasing over the stages. The line plots depict the CRs for each task, showing a gradual increase over the stages.

(hereafter referred to as the TVLL-CE algorithm), and perform 1000 Monte Carlo simulations. The simulations are conducted under two sets of scenarios.

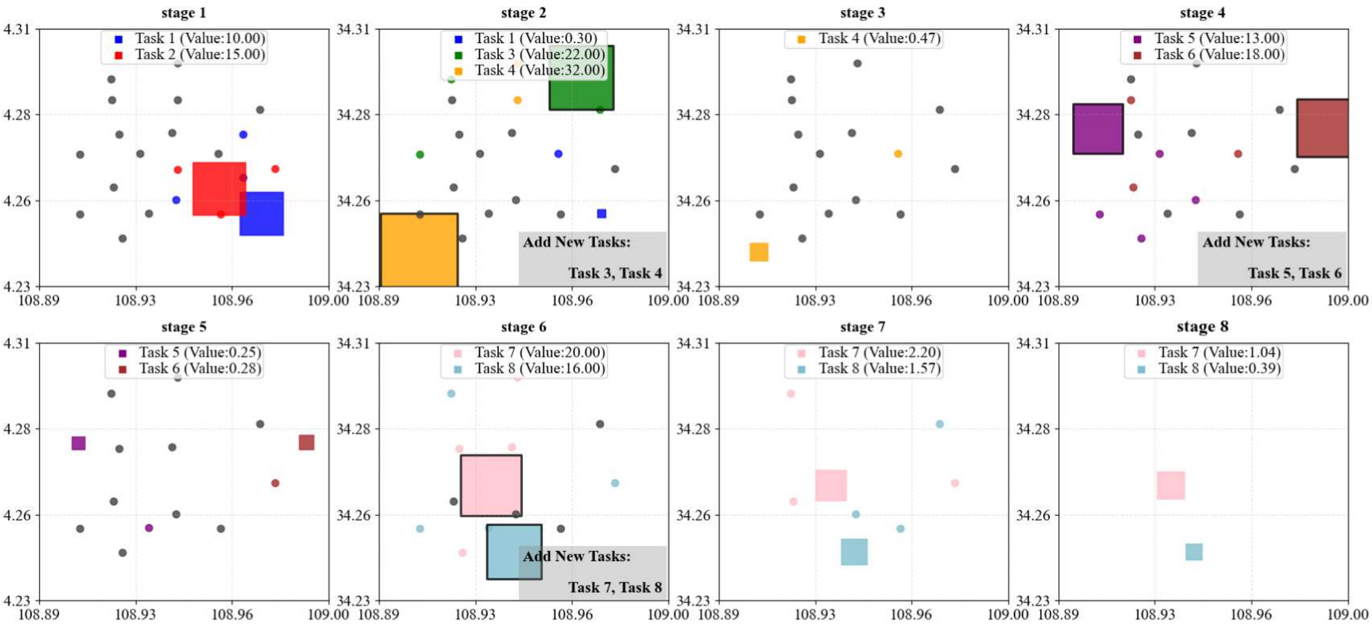

Fig. 15. Dynamic task assignment process with tasks added midway. The initial setup consists of 20 agents and two tasks. At stage 1, the agents are grouped into three coalitions, with a large number of agents not assigned to any tasks (indicated by gray circles). By stage 2, only task 1 remains. At this stage, tasks 3 and 4 are added, prompting a reallocation of tasks and the formation of four coalitions. This iterative process continues, with additional tasks added at stages 4 and 6. It is evident that the number of idle agents steadily decreases as tasks are gradually completed. Ultimately, after all agents' resources are exhausted, tasks 7 and 8 remain unfinished.

1) Fixed task quantity $m = 20$ with varying agent quantities $n \in \{100, 150, 200, 250, 300\}$.
2) Fixed agent quantity $n = 200$ with varying task quantities $m \in \{10, 15, 20, 25, 30\}$.

Figs. 8 and 9 present the statistical results as box plots.

In the left subfigure of Fig. 8, the global utility achieved by the ILLA algorithm is lower than that of TVLL-CE and ILLA-MACE. This is due to the absence of CE strategies, leading to local optima. ILLA-MACE achieves the highest utility, and the gap between it and TVLL-CE narrows as the number of agents increases, demonstrating its superior performance in resource-constrained scenarios.

In the right subfigure of Fig. 8, the running time of TVLL-CE increases notably with more agents due to exhaustive traversal, while ILLA-MACE maintains low and stable running time by selectively updating agents based on inertia. This result further validates the effectiveness of ILLA-MACE in enhancing global utility and maintaining computational efficiency across different agent scales.

Fig. 9 further demonstrates the superior performance of the ILLA-MACE algorithm over the TVLL-CE algorithm in terms of both average global utility and running time.

D. Performance Under Local Communication Scenario

In this section, we focus on a full communication scenario and compare our approach with the distributed mutex coalition formation (DMCF) algorithm [17] under the scenario involving 20 agents and six tasks to evaluate the effectiveness of the TB-ILLA-MACE algorithm.

In this section, we set the communication radius $r \in [r_{\min}, r_{\max}]$ in Fig. 10 to assess the effectiveness of our approach under various local communication scenarios.

Fig. 11 shows that the proposed TB-ILLA-MACE exhibits a faster convergence speed and achieves a higher utility value in scenarios with local communication compared to DMCF. Furthermore, TB-ILLA-MACE exhibits a slight gap in terms of utility compared to ILLA-MACE (full communication) at convergence.

In Fig. 12, global utility increases with the number of neighbors for both TB-ILLA-MACE and DMCF, underscoring the impact of communication structure. Notably, TB-ILLA-MACE shows higher global utility compared to DMCF, approaching the result in full communication. This demonstrates its effectiveness in local communication settings.

E. Performance Under Dynamic Task Assignment Scenario

In this section, we evaluate the effectiveness of our approach in dynamic task assignment scenarios under the resource-constrained scenario.

Fig. 13 illustrates the dynamic task allocation process under the given scenario.

In Fig. 14, stage 1 shows the initial task assignments among the agents. In stage 2 and stage 3, as the agents consume their resources to execute tasks, the value of all tasks decreases, and the size of the agent groups diminishes. In stage 4, tasks 1, 2, and 5 remain uncompleted, as all agents have exhausted their resources and no agents remain. Notably, the higher-value tasks are prioritized for completion, further validating the effectiveness of our algorithm.

Then, to evaluate the scalability of our approach, we investigate a dynamic task allocation scenario where new tasks are introduced midway while keeping the number of agents and their resources fixed.

As illustrated in Fig. 15, in scenarios with a sudden increase in the number of tasks, our approach is able to effectively

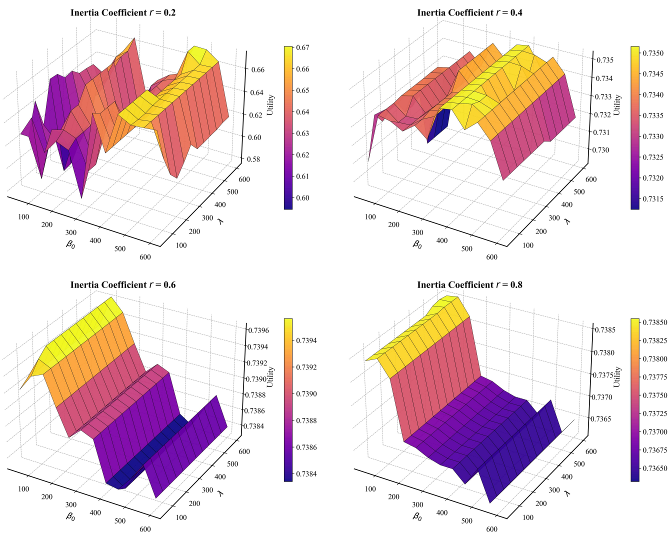

Fig. 16. Parameter sensitivity analysis of utility with respect to initial learning rate β_0 and decay factor λ. Each subplot corresponds to a fixed value of r, showing how utility varies across different combinations of β_0 and λ. Utility is relatively insensitive to λ, while sensitivity to β_0 varies with r.

complete tasks through coalition-based reallocation, thereby further validating its scalability.

F. Parameters Sensitivity Analysis

In this section, we investigate the sensitivity of the system utility to three critical hyperparameters: 1) the initial learning rate β_0; 2) the learning rate decay factor λ; and 3) the inertia coefficient r.

First, we select four sets of inertia coefficients. For each set, we let β and λ vary from 50 to 600 in increments of 50. Under each parameter combination, we conduct 100 Monte Carlo simulations under the same scenario settings as in Section VI-A, and record the average utility.

In Fig. 16, the utility surface remains relatively flat with respect to λ, indicating that the algorithm is sensitive to λ. Given λ, when the inertia coefficient is small ($r = 0.2$ and 0.4), the average utility fluctuates significantly with increasing β_0. In contrast, when the inertia coefficient is large ($r = 0.6$ and 0.8), the average utility exhibits a decreasing trend as β_0 increases. These results indicate a potential interaction between β_0 and r in influencing the algorithm's performance. Next, we conduct experiments to investigate their relationship.

In Fig. 17, the system utility exhibits sensitivity to the inertia coefficient r, under two representative learning rate configurations. When a low learning rate ($\beta_0 = 200$) is adopted, utility increases with r and peaks at $r = 0.65$. In contrast, a high learning rate ($\beta_0 = 500$) performs best at a lower inertia ($r = 0.35$), but its performance declines sharply as r increases.

As shown in Fig. 17, a smaller inertia coefficient leads to more agents participating in updates, which enhances exploration. In this case, a larger learning rate can improve overall utility. Conversely, a larger inertia coefficient limits participation and exploration. Consequently, a smaller learning rate is required to prevent the algorithm from converging prematurely to local optima.

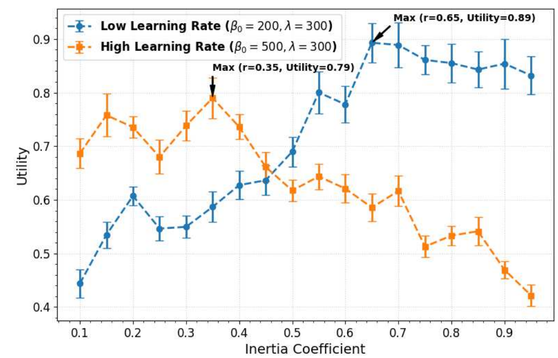

Fig. 17. Impact of inertia coefficient on utility under different learning rates. The blue curve represents the low learning rate setting ($\beta_0 = 200, \lambda = 300$), and the orange curve represents the high learning rate setting ($\beta_0 = 500, \lambda = 300$). Error bars indicate the standard error over multiple trials. The maximum utility for each setting is annotated on the plot.

VII. Conclusion

This article developed a CFG model for task allocation in heterogeneous multiagent systems, considering varying resource capacities and requirements. We show that the model constitutes a potential game with submodular coalitions and global utilities, improving the Nash equilibrium efficiency lower bound to $e/(2e-1)$. To find better Nash equilibria, ILLA-MACE is proposed to facilitate cooperative strategy exchange, enhancing global utility. To address local communication scenarios, the algorithm is extended as TB-ILLA-MACE. Finally, the effectiveness of our approach is validated through multiple simulations.

Future research will focus on extending the current theoretical framework to real-world applications and developing a dynamic task allocation model that optimizes long-term utility accumulation. This will enable farsighted decision-making, overcoming the limitations of short-sighted optimization.

References

[1] H. Wu and H. Shang, "Potential game for dynamic task allocation in multi-agent system," *ISA Trans.*, vol. 102, pp. 208–220, Jul. 2020.

[2] Y. Sheng, W. Tang, Q. Xiao, Z. Zeng, and N. R. Pal, "Leader-following consensus of time-scale-type heterogeneous nonlinear MASs via periodic event-triggered control," *IEEE Trans. Cybern.*, vol. 55, no. 9, pp. 4480–4490, Sep. 2025.

[3] H. Wang, H. Luo, Y. Jiang, and S. Wu, "Game-based distributed decision optimization for heterogeneous multiagent systems with unknown nonlinear dynamics," *IEEE Trans. Cybern.*, vol. 55, no. 9, pp. 4054–4063, Sep. 2025.

[4] Y. Wan, Y. Zhong, A. Ma, and L. Zhang, "An accurate UAV 3-D path planning method for disaster emergency response based on an improved multiobjective swarm intelligence algorithm," *IEEE Trans. Cybern.*, vol. 53, no. 4, pp. 2658–2671, Apr. 2023.

[5] P. Xue, L. Fei, and W. Ding, "A volunteer allocation optimization model in response to major natural disasters based on improved Dempster–Shafer theory," *Expert Syst. Appl.*, vol. 236, Feb. 2024, Art. no. 121285.

[6] J. Zhang, L. Ye, Z. Hou, L. Yu, and J. Cai, "Adaptive distributed cooperative tracking control and application for multiagent formation under communication constraints," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 4, pp. 4492–4506, Aug. 2024.

[7] F. Yan, J. Chu, J. Hu, and X. Zhu, "Cooperative task allocation with simultaneous arrival and resource constraint for multi-UAV using a genetic algorithm," *Expert Syst. Appl.*, vol. 245, Jul. 2024, Art. no. 123023.

[8] J.-J. Ji, Y.-N. Guo, X.-Z. Gao, D.-W. Gong, and Y.-P. Wang, "Q-learning-based hyperheuristic evolutionary algorithm for dynamic task allocation of crowdsensing," *IEEE Trans. Cybern.*, vol. 53, no. 4, pp. 2211–2224, Apr. 2023.

[9] X. Jia, H. Li, X. Chi, and B. Li, "Leader-following secure output consensus of heterogeneous multiagent systems based on two sampling mechanisms under hybrid cyber-attacks," *IEEE Trans. Cybern.*, vol. 54, no. 12, pp. 7826–7838, Dec. 2024.

[10] X. Wang, Z. Yin, Y. Lei, T. Huang, and J. Kurths, "Secure consensus for switched multiagent systems under DoS attacks: Hybrid event-triggered and impulsive control approach," *IEEE Trans. Cybern.*, vol. 55, no. 5, pp. 2400–2410, May 2025.

[11] S. Xiao, X. Ge, Q.-L. Han, and Y. Zhang, "Secure distributed adaptive platooning control of automated vehicles over vehicular ad-hoc networks under denial-of-service attacks," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 12003–12015, Nov. 2022.

[12] D. Xu et al., "An efficient dynamic resource allocation framework for evolutionary bilevel optimization," *IEEE Trans. Cybern.*, vol. 55, no. 2, pp. 726–739, Nov. 2024.

[13] E. Czarnecki and A. Dutta, "Scalable hedonic coalition formation for task allocation with heterogeneous robots," *Intell. Service Robot.*, vol. 14, no. 3, pp. 501–517, Jun. 2021.

[14] J. Xiong, J. Li, J. Li, S. Kang, C. Liu, and C. Yang, "Probability-tuned market-based allocations for UAV swarms under unreliable observations," *IEEE Trans. Cybern.*, vol. 53, no. 11, pp. 6803–6814, Nov. 2023.

[15] H.-L. Choi, L. Brunet, and J. P. How, "Consensus-based decentralized auctions for robust task allocation," *IEEE Trans. Robot.*, vol. 25, no. 4, pp. 912–926, Aug. 2009.

[16] N. Edalat, C.-K. Tham, and W. Xiao, "An auction-based strategy for distributed task allocation in wireless sensor networks," *Comput. Commun.*, vol. 35, no. 8, pp. 916–928, May 2012.

[17] I. Jang, H.-S. Shin, and A. Tsourdos, "Anonymous hedonic game for task allocation in a large-scale multiple agent system," *IEEE Trans. Robot.*, vol. 34, no. 6, pp. 1534–1548, Dec. 2018.

[18] J. Hu, P. Bhowmick, I. Jang, F. Arvin, and A. Lanzon, "A decentralized cluster formation containment framework for multirobot systems," *IEEE Trans. Robot.*, vol. 37, no. 6, pp. 1936–1955, Dec. 2021.

[19] L. Wang, T. Qiu, Z. Pu, J. Yi, J. Zhu, and W. Yuan, "Hedonic coalition formation for distributed task allocation in heterogeneous multi-agent system," *Int. J. Control, Autom. Syst.*, vol. 22, no. 4, pp. 1212–1224, Feb. 2024.

[20] W. Saad, Z. Han, T. Basar, M. Debbah, and A. Hjorungnes, "Hedonic coalition formation for distributed task allocation among wireless agents," *IEEE Trans. Mobile Comput.*, vol. 10, no. 9, pp. 1327–1344, Sep. 2011.

[21] W. Yang, X. Liu, L. He, Y. Du, B. Q. Vo, and Y. Chen, "Game-theoretic distributed approach for heterogeneous-cost task allocation with budget constraints," *Expert Syst. Appl.*, vol. 255, Dec. 2024, Art. no. 124721.

[22] T. Zhang, Y. Wang, Z. Ma, and L. Kong, "Task assignment in UAV-enabled front jammer swarm: A coalition formation game approach," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 6, pp. 9562–9575, Dec. 2023.

[23] N. Qi, Z. Huang, F. Zhou, Q. Shi, Q. Wu, and M. Xiao, "A task-driven sequential overlapping coalition formation game for resource allocation in heterogeneous UAV networks," *IEEE Trans. Mobile Comput.*, vol. 22, no. 8, pp. 4439–4455, Aug. 2023.

[24] J. Chen et al., "Joint task assignment and spectrum allocation in heterogeneous UAV communication networks: A coalition formation game-theoretic approach," *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 440–452, Jan. 2021.

[25] K. Xiong, T. Zhang, G. Cui, S. Wang, and L. Kong, "Coalition game of radar network for multitarget tracking via model-based multiagent reinforcement learning," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 3, pp. 2123–2140, Jun. 2023.

[26] K. Xiong, G. Cui, M. Liao, S. Li, and L. Kong, "Hierarchical task and power coalition game of radars for maneuvering target tracking in self-defensive suppressive jamming," *IEEE Trans. Veh. Technol.*, vol. 74, no. 2, pp. 3238–3252, Feb. 2025.

[27] J. R. Marden, G. Arslan, and J. S. Shamma, "Joint strategy fictitious play with inertia for potential games," *IEEE Trans. Autom. Control*, vol. 54, no. 2, pp. 208–220, Feb. 2009.

[28] Y. Yazıcıoğlu, R. Bhat, and D. Aksaray, "Distributed planning for serving cooperative tasks with time windows: A game theoretic approach," *J. Intell. Robotic Syst.*, vol. 103, no. 2, p. 27, Sep. 2021.

[29] H. P. Young, *Strategic Learning and Its Limits*. London, U.K.: Oxford Univ. Press, 2004.

[30] C. Sun, "A time variant log-linear learning approach to the SET K-COVER problem in wireless sensor networks," *IEEE Trans. Cybern.*, vol. 48, no. 4, pp. 1316–1325, Apr. 2018.

[31] G. Arslan, J. R. Marden, and J. S. Shamma, "Autonomous vehicle-target assignment: A game-theoretical formulation," *J. Dyn. Syst., Meas., Control*, vol. 129, no. 5, pp. 584–596, Sep. 2007.

[32] D. H. Wolpert and K. Tumer, "An introduction to collective intelligence," 1999, *arXiv:cs/9908014*.

[33] A. Vetta, "Nash equilibria in competitive societies, with applications to facility location, traffic routing and auctions," in *Proc. 43rd Annu. IEEE Symp. Found. Comput. Sci.*, Nov. 2002, pp. 416–425.

[34] T. H. Ho, C. F. Camerer, and J.-K. Chong, "Self-tuning experience weighted attraction learning in games," *J. Econ. Theory*, vol. 133, no. 1, pp. 177–198, Mar. 2007.

[35] L. E. Blume, "The statistical mechanics of strategic interaction," *Games Econ. Behav.*, vol. 5, no. 3, pp. 387–424, Jul. 1993.

[36] J. R. Marden and J. S. Shamma, "Revisiting log-linear learning: Asynchrony, completeness and payoff-based implementation," *Games Econ. Behav.*, vol. 75, no. 2, pp. 788–808, Jul. 2012.

[37] A. Maddux, R. Ouhamma, and M. Kamgarpour, "Finite-time convergence to an ϵ-efficient Nash equilibrium in potential games," 2024, *arXiv:2405.15497*.

[38] Z. Li, C. Liu, S. Tan, and Y. Liu, "A timestamp-based log-linear algorithm for solving locally-informed multi-agent finite games," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123677.

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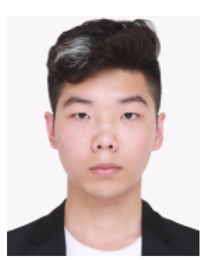

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